

Engineering Physics (2025)

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Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

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p-n junction diode – forward and reverse conditions

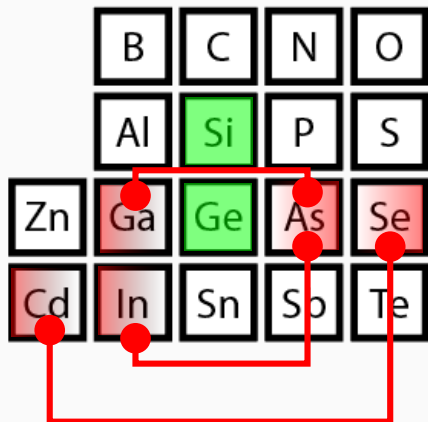
Solar cell

Extrinsic semiconductor

Periodic table

 *Elemental semiconductor*

 *Compound semiconductor*



	B	C	N	O
	Al	Si	P	S
Zn	Ga	Ge	As	Se
Cd	In	Sn	Sb	Te

Elemental	Compound
Si	GaAs
Ge	InAs
	CdSe

Table 1: Intrinsic semiconductors.

Extrinsic semiconductor – Dopant atoms

Definition

- Extrinsic semiconductor is a semiconductor with impurities added.
- Depending upon the type of impurities added, extrinsic semiconductor is of two types – n-type and p-type

n-type	p-type
P	Al
As	B
Sb	Ga
In	

Table 2: Dopants for elemental s.c.s.

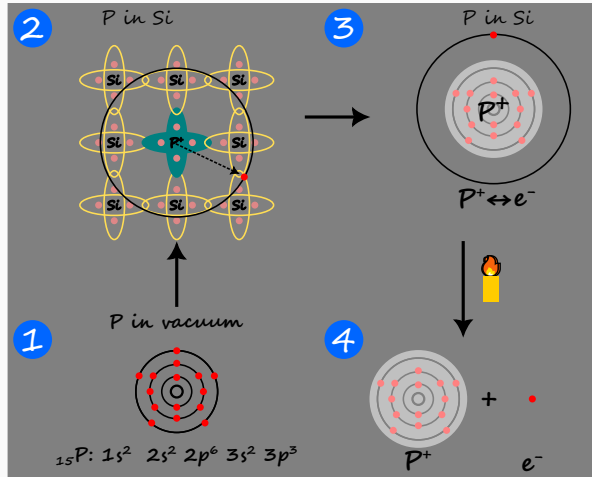
n-type	p-type
Se	Zn
Te	Cd
Si at Ga	Si at As
Ge at Ga	Ge at As

Table 3: Dopants for compound s.c.s.

Problem

Prove that Si doped at Ga position in GaAs acts as a donor dopant.

Donor impurity – Hydrogen-like atom



1. P as isolated atom.
2. P as impurity inside Si.
3. P impurity as $\text{P}^+ \leftrightarrow e^-$ hydrogen like atom.
4. Due to thermal energy, ionization of P impurity gives free electron.

Key Insight

Doping with donor impurity forms an n-type semiconductor.



Donor impurity – Hydrogen-like atom

Definition

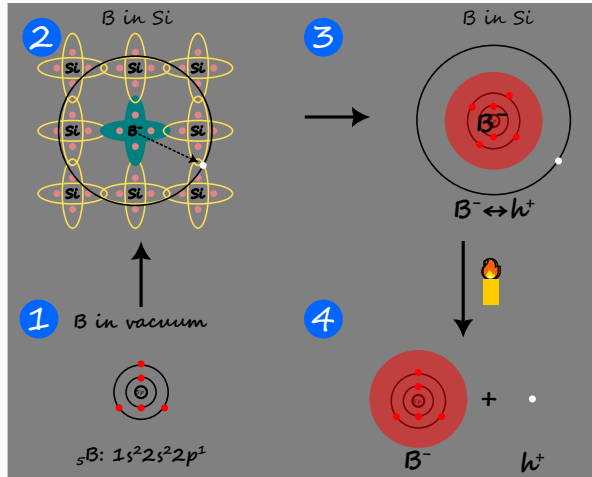
- Donor impurity, also called donor dopant, is an atom that donates an electron to the intrinsic semiconductor upon ionization.
1. Phosphorus in vacuum has 5 electrons in the valence shell.
 2. Phosphorus in silicon shares 4 electrons with neighbouring Si atoms for covalent bonding. The **extra** electron is loosely coupled to the phosphorus nucleus.
 3. At absolute zero temperature, phosphorus can be modeled as a hydrogenic atom with nucleus as phosphorus ion of charge +1, and a **bound** electron of charge -1 revolving around the nucleus.



4. When the thermal energy is sufficient to overcome the attractive force between P^+ and e^- , the electron becomes **free** and is released into the Si lattice.



Acceptor impurity – Hydrogen-like atom



Key Insight

Doping with acceptor impurity forms a p-type semiconductor.



Acceptor impurity – Hydrogen-like atom

Definition

- Acceptor impurity, also called acceptor dopant, is an atom that accepts an electron from the intrinsic semiconductor upon ionization.
1. Boron in vacuum has 3 electrons in the valence shell.
 2. Boron in silicon shares 3 electrons with neighbouring Si atoms for covalent bonding. The 4th electron is absent and can be considered as an **extra** hole that is loosely coupled to the boron nucleus.
 3. At absolute zero temperature, Boron can be modeled as a hydrogenic atom with nucleus as Boron ion of charge -1, and a **bound** hole of charge +1 revolving around the nucleus.

$$\text{B in Si} \equiv \text{B}^- \leftrightarrow \text{h}^+$$

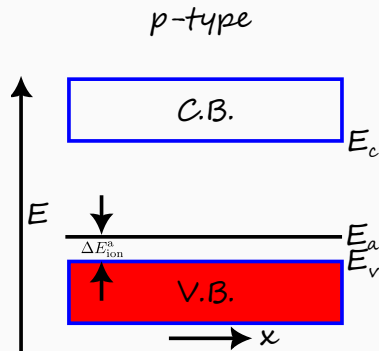
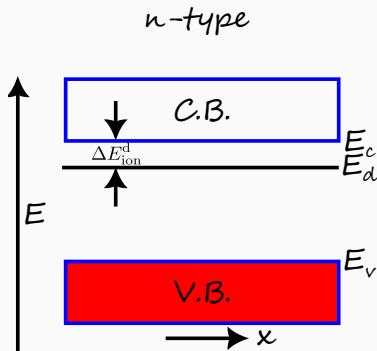
4. When the thermal energy is sufficient to overcome the attractive force between B^- and h^+ , the hole becomes **free** and is released into the Si lattice.

$$\text{B}^- \leftrightarrow \text{h}^+ + \mathcal{O}(k_B T) \rightarrow \text{B}^- + \text{h}^+$$

Dopant energy levels

Definition

- Donor energy level E_d is the ground state of donor dopant atom.
- Acceptor energy level E_a is the ground state of acceptor dopant atom.



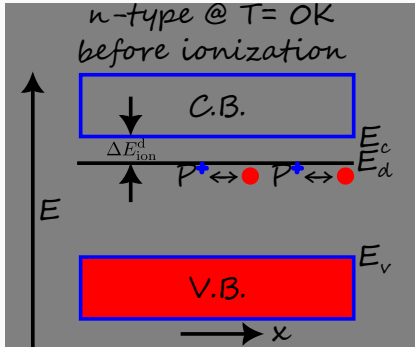
Key Insight

Donor energy level is closer to the conduction band minimum and acceptor energy is closer to the valence band maximum.

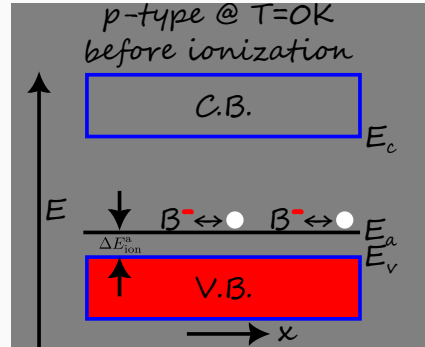


Dopant energy levels – before ionization

- At absolute zero temperature, due to lack of thermal energy, the dopant atoms are **not** ionized.



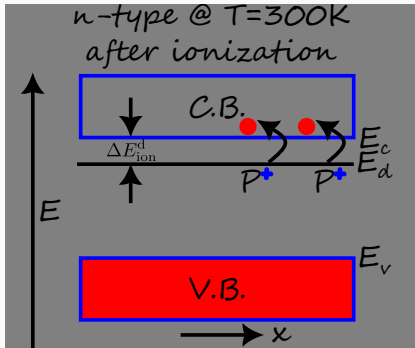
- The electron is **bound** to the donor ion.



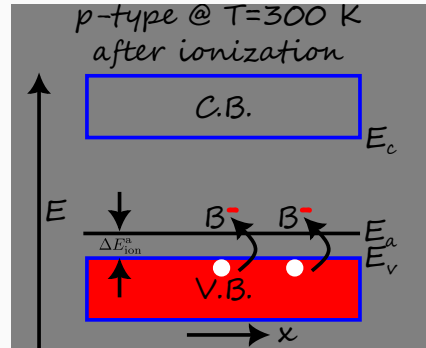
- The hole is **bound** to the acceptor ion.

Dopant energy levels – after ionization

- At room temperature, the thermal energy is of the order of ionization energy. This leads to ionization of dopant atoms.



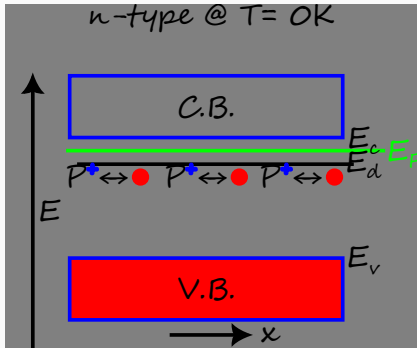
- Upon ionization, the donor dopant releases free electron into the conduction band.



- Upon ionization, the acceptor dopant accepts electron from valence band and creates a hole in the valence band.

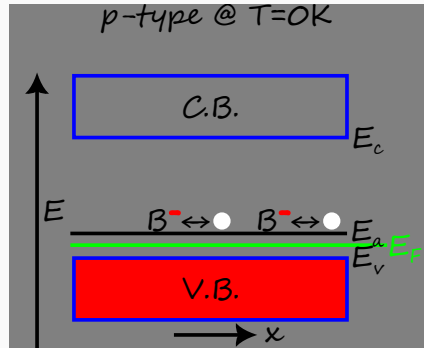
Position of Fermi level – @ 0 K

- At absolute zero temperature, all the energy levels below the Fermi level must be **occupied** by electrons.



The Fermi energy level lies

- above** the donor energy level.
- below** the conduction band edge.

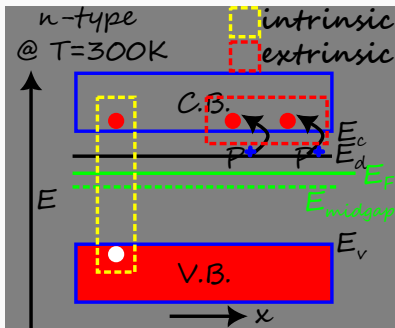


The Fermi energy level lies

- below** the acceptor energy level.
- above** the valence band edge.

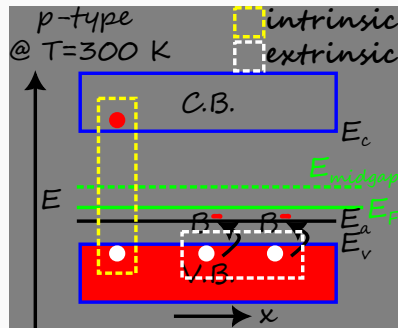
Position of Fermi level – @ 300 K

- At room temperature, there is contribution to the free carriers from both **extrinsic** and **intrinsic**



The Fermi energy level lies

- below** the donor energy level since donor energy level is **empty**.
- above** the midgap energy level.

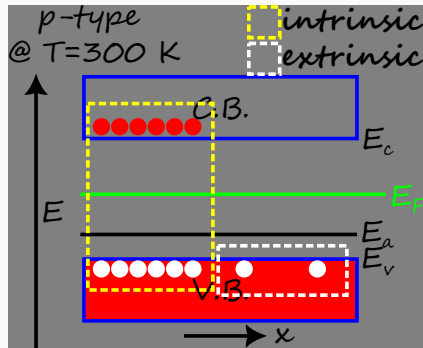
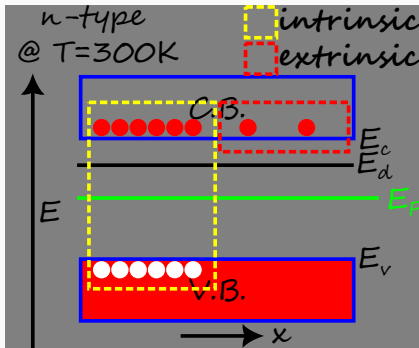


The Fermi energy level lies

- above** the acceptor energy level since acceptor energy level is **filled**.
- below** the midgap energy level.

Position of Fermi level – @ $T \gg 300\text{ K}$

- At very high temperature, the contribution to the free carriers from **intrinsic** dominates the contribution from **extrinsic**



- The Fermi energy level lies near the midgap energy level in both cases.

Key Insight

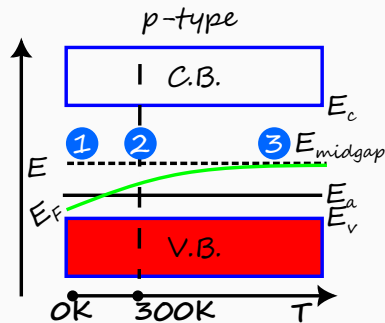
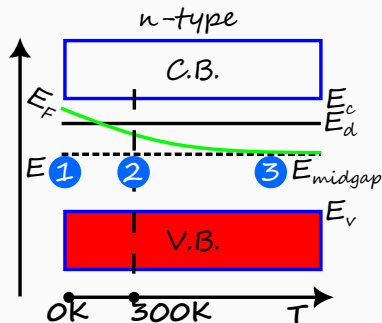
At very high temperature, extrinsic semiconductor behaves nearly as an intrinsic semiconductor.



Variation of Fermi level with temperature

Definition

- **Freeze-out** is the condition when the dopants are not ionized.
- **Complete ionization** is the condition when the dopants are completely ionized.



1. At 0 K, the dopants are under freeze-out condition.
2. At 300 K, the thermal energy is nearly equal to ionization energy of dopants leading to complete ionization condition.
3. At $T \gg 300$ K, the extrinsic s.c. behaves as an intrinsic s.c.

Test Your Understanding

1. What is a donor impurity? What is an acceptor impurity?
2. Differentiate between n-type and p-type semiconductor by sketching their band diagrams.
3. p-type and n-type have more holes and electrons respectively. Explain why they are electrically neutral.
4. What is meant by complete ionization? What is meant by freeze-out?
5. What is the effect of temperature on the working of n-type and p-type semiconductors? Draw energy band diagrams to show the variation in the position of Fermi level with rise in temperature.
6. Draw energy band diagrams for n-type semiconductor at 0 K and at room temperature.
7. Distinguish between extrinsic and intrinsic semiconductors.
8. Sketch a graph of n_0 versus temperature for an n-type semiconductor.
9. Sketch a graph of E_F versus temperature for p-type semiconductor.
10. What is the relevance of the Fermi energy?

- acceptor atoms
- donor atoms
- extrinsic semiconductor
- complete ionization
- freeze-out

End of M2U1

Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

pn junction at zero bias

Definition

Zero bias condition is considered when external battery is not connected across p-region and n-region.

- The word voltage bias is used in the context of electricity to refer to external force.
- At zero voltage bias, there is **no application of external force** on the charge carriers.
- However, there are two internal forces at play –
 1. Diffusion forces due to gradients in concentration of charge carriers.
 2. Drift forces due to internal electric fields.
- The interplay of these internal forces leads to net zero force on charge carriers at thermal equilibrium.

Key Insight

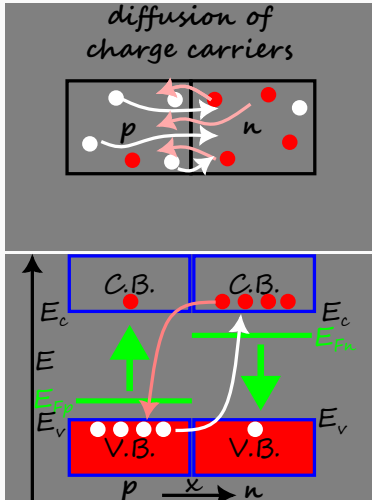
At zero bias, there is interplay of internal drift and diffusion forces.



pn junction – Diffusion

Definition

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.

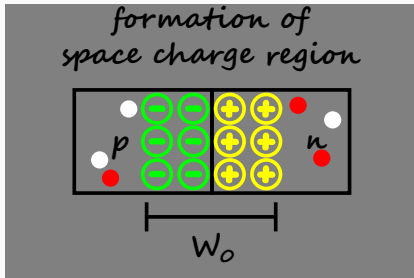


- The difference in Fermi level acts as a “driving force” so that electrons from n-region **diffuse** into the p-region.
- By similar analysis, holes from p-region diffuse into the n-region.
- Diffusion of electrons from n-region lowers its Fermi level E_{Fn} .
- Diffusion of holes from p-region raises its Fermi level E_{Fp} .

pn junction at zero bias – Space-charge region

Definition

- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0 .



- Due to the diffusion of mobile charge carriers, their **ionic counterparts** are left without partners.
- Space charge region is also called the **depletion region** since the region is depleted ^a of the free charge carriers.

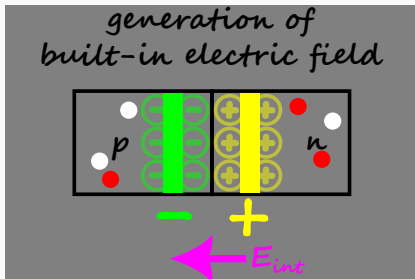
^adeplete means decrease or remove.

- The depletion region consists of positively charged donor ions on the n-side of the junction and negatively charged acceptor ions on the p-side of the junction.

pn junction at zero bias – Built-in electric field

Definition

Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.



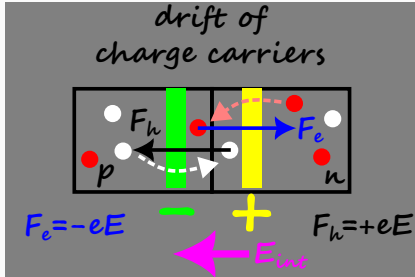
- Within the depletion region, the oppositely charged ions on either side of junction can be considered as plates of a capacitor.
- Since the plates are charged, a **built-in electric field** is generated.
- The built-in electric field is directed from **n-region to p-region**.

- E_{int} prevents further build up of space charge due to diffusion as it opposes the diffusion of electrons from n-region into p-region and vice versa.

pn junction at zero bias – Drift

Definition

Drift is the net movement of charged particles due to the application of electric field.



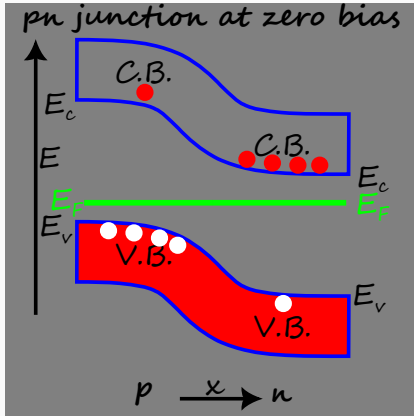
- Consider a diffused electron that is present in the p side of the depletion region.
- Since the built-in electric field is directed from n-region to p-region, the electric force is directed from p-region to n-region.
- Electron drift backs to the n-region.

- The diffusion force is counterbalanced by the drift force.
- Similar analysis applies to the diffused hole.

Problem

Prove the counterbalancing effect of drift and diffusion for holes.

pn junction at zero bias – Thermal equilibrium



Definition

Thermal equilibrium occurs when the net force on the system is zero.

- Since the difference in Fermi level is a “driving” force, when the system reaches thermal equilibrium, the net force is zero.
- This implies the Fermi level is a constant throughout the system at thermal equilibrium.

- Since the external force is zero at zero bias, the pn junction is at thermal equilibrium.

Key Insight

At thermal equilibrium, E_F is a constant.



pn junction at non-zero bias

pn junction – at non-zero bias

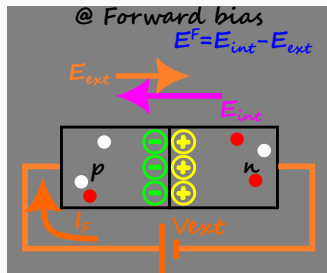
- At non-zero voltage bias, there is **an application of external force** on the charge carriers. As a result, the pn junction cannot attain thermal equilibrium.
- Since there is no thermal equilibrium, the Fermi energy is not a constant across the pn junction.
- Since the direction of built-in electric field is from n-region to p-region, the behaviour of p-n junction is different when the external field E_{ext} is parallel or anti-parallel to the built-in electric field E_{int} .
 1. $E_{\text{ext}} \uparrow \downarrow E_{\text{int}}$ corresponds to forward biasing.
 2. $E_{\text{ext}} \uparrow \uparrow E_{\text{int}}$ corresponds to reverse biasing.

Key Insight

At non zero bias, the Fermi energy is not a constant across the pn junction.



pn junction – at forward bias



Forward

$$V_F > 0$$

$$E^F < E_{\text{int}}$$

$$W_F < W_0$$

$$V_F < V_{\text{int}}$$

$$E_{F_n} - E_{F_p} > 0$$

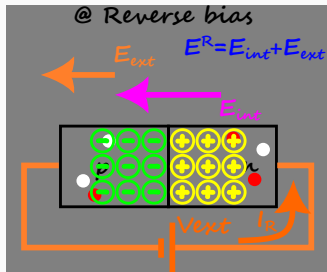
$$I_F > 0$$

Definition

Forward bias condition is considered when the positive electrode of battery with potential V_F is connected to p-region and negative electrode is connected to n-region.

- The external bias $V_{\text{ext}} = V_F$ is from p-region to n-region.
- The external field E_{ext} is from p-region to n-region. The internal field E_{int} is **anti-parallel** to E_{ext} . The net electric field E^F is lower than the case of zero bias.
- So less space charge is created in depletion region. So we have smaller space charge width W_F .

pn junction – at reverse bias



Reverse

$$V_R < 0$$

$$E^R > E_{\text{int}}$$

$$V_R > V_{\text{int}}$$

$$E_{Fn} - E_{Fp} < 0$$

$$W_R > W_0$$

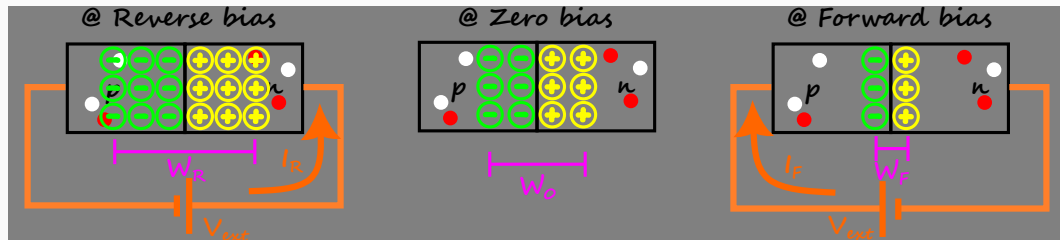
$$I_R < 0$$

Definition

Reverse bias condition is considered when the positive electrode of battery is connected to n-region and negative electrode is connected to p-region.

- The external bias $V_{\text{ext}} = V_R$ is from n-region to p-region.
- The external field is from n-region to p-region.
Under reverse bias condition, the internal field is **parallel** to the external field. The net electric field E^R is higher than the case of zero bias.
- So more space charge is created in depletion region. So we have larger space charge width W_R .

Comparison of depletion region widths at different biases

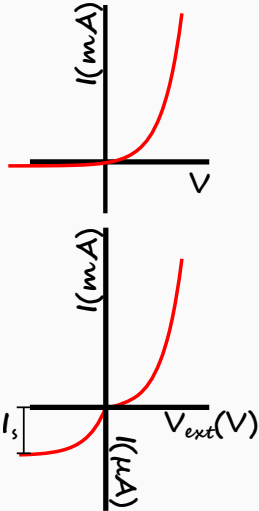


$$W_R > W_0 > W_F$$

Large forward current vs tiny reverse current

- Under biasing conditions, the electrodes of the battery change the Fermi levels of the p and n regions.
- Since the negative electrode supplies electrons, it raises the Fermi level of the connected region. Similarly, positive electrode lowers the Fermi level.
- Under forward biasing, the energy of electrons at n-region is raised. The flow of electrons to lower energy p-region is possible. Similarly, the energy of holes at p-region is lowered. The flow of holes to higher energy n-region is possible. This leads to **large forward current**.
- Under reverse biasing, the energy of electrons at n-region is lowered. The flow of electrons to higher energy p-region is difficult. Similarly, the energy of holes at p-region is raised. The flow of holes to lower energy p-region is difficult.
- However, the reverse bias is favourable for minority electrons in p region and minority holes in n-region. Since the number of these carriers is very few, we see a **tiny reverse current**.

pn junction – $I - V$ characteristics



Definition

Rectifying action of pn junction is its property to allow **uni-directional** conduction.

- The diode equation can be shown to be

$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right]$$

where I_s is the reverse saturation current.

- The equation and the $I - V$ capture the non-linear nature of p-n junction.
- In the forward bias condition, it acts as a “closed switch” allowing large current to flow through it.
- In the reverse bias condition, it acts as an “open switch” allowing insignificantly low current to flow through it.
- This unidirectional conduction is called rectifying action.

Conceptual Questions

Problem

What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined?

[Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]

Tough problems

Problem

Prove that the analysis of pn junction using potential energy barrier maps with the analysis using Fermi energy level. In other words, prove that $E_{Fn} - E_{Fp} = eV_{\text{ext}}$.

Test Your Understanding

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region. What is the typical value of depletion region width for commonly used pn junctions?
4. Draw band diagrams for symmetrically doped pn junction at
 - zero bias
 - forward bias
 - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

Technical terms

- metallurgical junction
- diffusion
- built-in electric field
- thermal equilibrium
- built-in potential barrier
- depletion region
- space charge region
- space charge width
- reverse saturation current
- reverse breakdown

Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

Optoelectronics

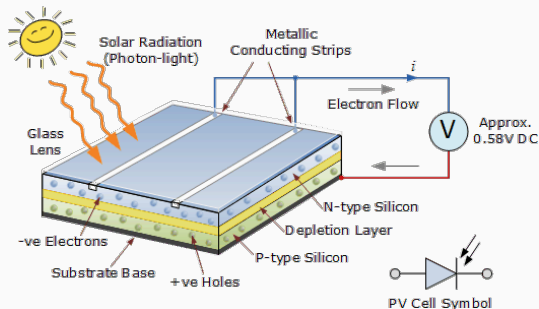
- Electronics is the study of controlling electrical signals.
- Any electrical signal is characterized by its current and voltage properties.
- Electronics is broadly classified into three areas –
 - **Microelectronics** – here we study switching electrical signals, which is useful to represent binary logic. This is necessary for computation. This is also called **digital electronics**. Since power is not a concern, signals have currents of the order of microamperes (hence the name microelectronics) and voltages of the order of few volts
 - **Power electronics** – here we study amplifying electrical signals to transfer energy. This is necessary for energy requirements of society as it is well known that electrical energy is efficient way of energy transfer. Signals have currents of the order of hundreds of amperes and voltages of the order of thousands of volts. This is also called **analog electronics**.
 - Opto-electronics ¹ – here we harness the interaction between electron and photon. The advantages of light like its high velocity, free source of energy (Sun light), are combined with the advantages of electron like its easy manipulation to develop devices of specific requirements.

¹Opto-electronics is a combination of optics and electronics

Solar cell – construction

Definition

- Solar cell, also technically called the **photovoltaic cell** is an opto-electronic device based on pn junction that can generate electrical power when illuminated.
- A collection of cells in series and parallel combination is called a solar panel



Sketch credit: Alternative Energy Tutorials

Sketch of solar panel. Sketch taken from Alternative Energy Tutorials

- The main aim of solar cell operation is to convert solar energy into electrical energy.
- Electrodes are deposited on both sides of junction. The top metal electrode is in the form of metal grid with fingers, which permits sunlight to pass through it.

Solar cell – working principle

Definition

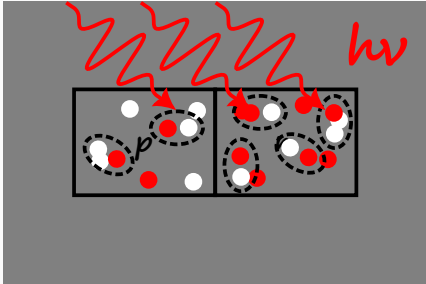
The working principle of solar cell involves three processes–

1. Generation of **excess** electron-hole pairs upon the illumination of light, and
2. Separation of electrons from holes **before** radiative recombination due to the built-in electric field.
3. Generation of voltage due to **accumulation** of excess separated charge carriers at the ends of the pn junction
 - The solar cell pn junction is operated without bias i.e. zero bias.
 - Upon illumination with photons of suitable energy, the photon are absorbed through the surface area. The covalent bonds of the lattice are broken releasing the free electrons and holes into the conduction band and valence band respectively.

Problem

A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?

Working principle – 1. Excess electron-hole pairs



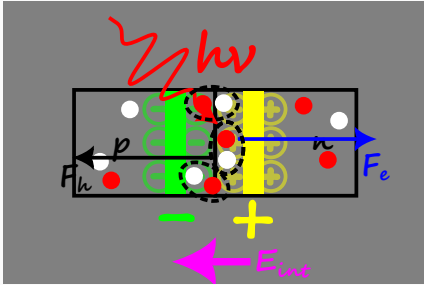
- Upon illumination with photons with energy greater than the band gap, optical absorption takes place.
- This results in the generation of excess electron-hole pairs throughout the device.

Key Insight

Optical absorption generates excess electron-hole pairs.



Working principle – 2b. Illumination at depletion region



- At the depletion region also, the excess charge carriers are generated.
- However, before they undergo radiative recombination, the built-in electric field sweeps them out of the depletion.

Key Insight

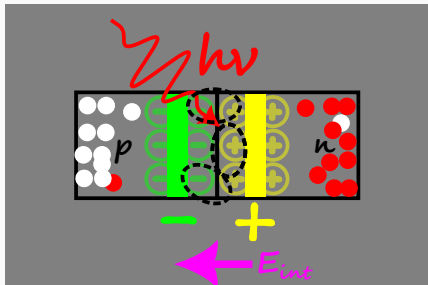
At the depletion region, separation of excess charge carriers prevents radiative recombination.



Working principle – 3. Photo-voltage

Definition

Photovoltage, also called photo e.m.f.² is the voltage appearing across the device due to the separation of excess charge carriers.



- The swept excess electrons and holes **accumulate** at the n region and p region edges respectively
- The accumulation of positive charge at the p-edge and negative charge at the n-edge leads to the generation of a voltage V_L .

- Since the source of voltage is illumination of photons, it is called **photovoltage**.

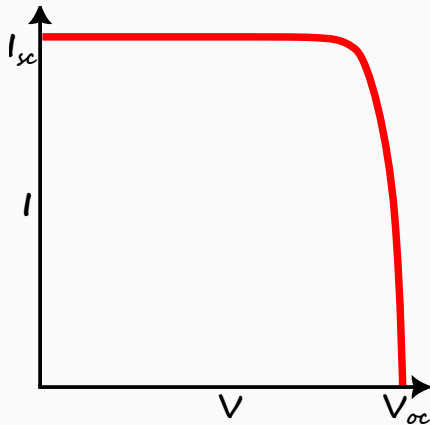
Key Insight

The photovoltage forward biases the pn junction.



²e.m.f. stands for electro motive force

Solar cell – $I - V$ characteristics



- From Kirchoff voltage law, the photo-voltage is given by

$$V_L = I_L R_L$$

- From diode $I - V$ characteristics, the forward current I_F is given by

$$I_F = I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right]$$

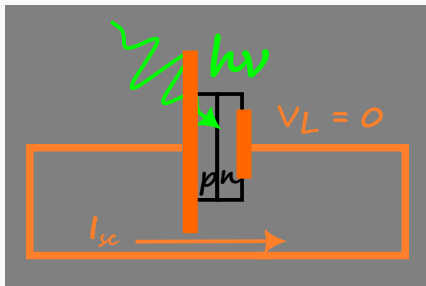
- Therefore the solar cell $I - V$ characteristics is given by

$$\begin{aligned} I_{\text{net}} &= I_L - I_F \\ &= I_L - I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right] \end{aligned}$$

Solar cell – short circuit current

Definition

Short circuit current is defined the current through solar cell when it is short circuited or zero load resistance is connected.



- When the load resistance is zero, the solar cell generates maximum current. This condition is called short circuit condition.
- At short circuit condition, $R_L = 0$, so the voltage drop across load is zero

$$V_L = I_L R_L = 0$$

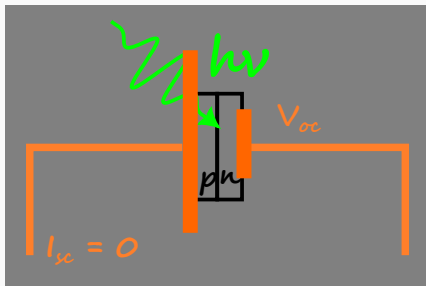
- The net current is the short circuit current I_{sc} and is given by

$$\begin{aligned} I_{sc} &= I_L - I_s (e^0 - 1) \\ &= I_L \end{aligned}$$

Solar cell – open circuit voltage

Definition

Open circuit voltage is defined the voltage across solar cell when it is open circuited or infinite load resistance is connected.



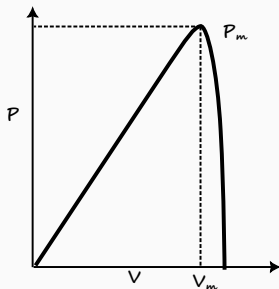
- When the load resistance is infinite, the solar cell generates maximum voltage. This condition is called open circuit condition.
- At open circuit condition, $R_L = \infty$, so the current is zero

$$I_L = \frac{V_L}{R} = 0$$

- The photovoltage is the open circuit voltage V_{oc} and is given by

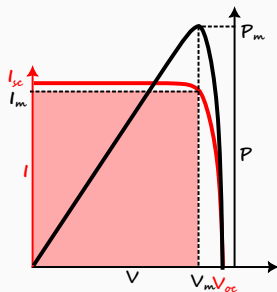
$$I_L - I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right] = 0 \Rightarrow I_L = I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right]$$
$$\exp \left(\frac{eV_{oc}}{k_B T} \right) = \frac{I_L}{I_s} + 1 \Rightarrow V_{oc} = \left(\frac{k_B T}{e} \right) \ln \left(1 + \frac{I_L}{I_s} \right)$$

Solar cell – $P - V$ characteristics



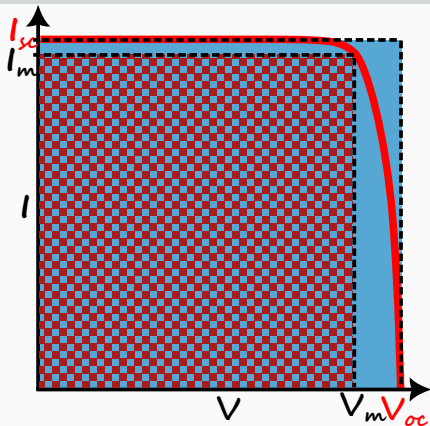
- The power generated in solar cell is completely consumed in the load resistance.
- If the photocurrent at V_m is I_m then the maximum power is

$$P_m = V_m I_m$$



- In $P - V$ curve, this is represented as (V_m, P_m) .
- The I & $P - V$ curves show the mappings of $I - V$ and $P - V$ curves. In $P - V$ curve, maximum power is represented as as red rectangle.

Output electrical power – Fill factor



- From the $I - V$ characteristics of solar cell, it is evident that the maximum current generated cannot exceed the short circuit current I_{sc} and the maximum voltage generated cannot exceed the open circuit voltage V_{oc} .
- The theoretically possible maximum power that can be generated should be less than the product of I_{sc} and V_{oc} . In the $I - V$ curve, this is represented as blue rectangle.

- The ratio of experimentally possible maximum power to theoretically possible maximum power is called Fill Factor and is given by

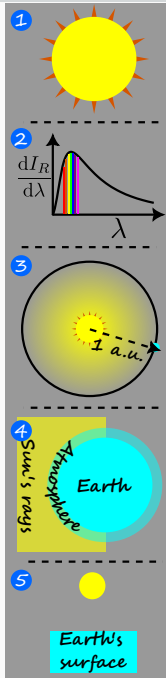
$$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$$

Key Insight

Fill factor is a material property.



Input optical power – Solar irradiance



Definition

Solar irradiance is the solar power incident on the Earth's surface per unit area.

1. Sun as blackbody – Since sun is the major source of light energy, the input power is determined by the properties of sun.
2. Planck's blackbody radiation law – Sun is emitting photons of various wavelengths per unit time. Each photon with a given wavelength has an energy given by the Planck's law.
3. Sun -Earth separation – However, we are only receiving a fraction of the Sun's power.
4. Solar irradiance outside atmosphere – It is 1360 W m^{-2} .
5. Solar irradiance at Earth's surface – It is nearly 1000 W m^{-2} at noon due to losses in atmosphere.

Practical considerations – Solar cell efficiency

Definition

Conversion efficiency is defined as ratio of output electrical power to input optical power.

- To harness solar energy, the following practical limitations need to be considered.
- All the power cannot be harnessed as the optical absorption by a solar cell has a cut off wavelength due to the energy band gap.
- The photons with energy greater than energy band gap also generate electron-hole pairs. However, the difference of photon energy and band gap is dissipated as heat.
- The sum total of all these considerations is that a fraction of incident optical power is converted into electrical power and is given by the conversion efficiency η as

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100$$

Solar cell – applications

Chandrayaan 2 mission at glance [Click for video]



Image credits: ISRO

Problem

From the data in above figure, estimate the conversion efficiency of material used in Pragyan rover.

Test Your Understanding

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.
3. A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?
4. An alien ship crashlanded onto Earth's surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun.

- optical absorption
- generation of excess charge carriers
- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current